

THE BALANCE OF REPRESENTATION AND COMPUTATION IN INTERFACE

1. Introduction

(1) purpose

a. **Interface Dualism**

since SPE, higher modules (morpho-syntax, maybe semantics) are assumed to have two distinct means of talking to phonology:

1. **procedurally**: the phonological cycle (and its later incarnations)
==> nothing is sent down
2. **representationally**: boundaries (and their later incarnations)
==> something is sent down

- b. I call an "**interface phenomenon**" a phonological phenomenon that has an extra-phonological conditioning.

Various interface theories have accounted for the same pool of interface phenomena with changing combinations of procedural and representational mechanisms.

In actual fact, a salient ID of an interface theory is the distribution of procedural and representational solutions over interface phenomena.

- c. therefore, it is important to have an **exhaustive overview of interface phenomena**:

1. what is out there?
2. how can interface phenomena be classified on theory-neutral grounds in such a way that any given phenomenon belongs to one and only one category?

I submit that the classification of Kenstowicz & Kisseberth (1977:83ss, 1979:407ss) is functional and meaningful.

==> I call this the **KK-classification**.

- d. such a classification is a prerequisite for assessing and comparing theories: "here are the classes of phenomena that every interface theory must be able to handle, so tell me how yours covers them."
- e. up to now, every interface theory tends to have its typical phenomena, on which it does well. Interface theories usually do not talk about the typical phenomena of the other theories.

This does not allow for a reasonable competition.

- f. existing interface theories will be scanned as to their behaviour in regard of the classification proposed.
- g. given the KK-classification, the association of
 1. Direct Interface (the interface theory that I am promoting)
 2. CVCV (the phonological theory that I am working in)shows a distribution of the phenomena over procedural and representational means that is different from all other interface theories.
- h. an interesting generalisation emerges:

stress-related phenomena = due to procedural intervention
segmental effects = due to representational intervention

I show that this makes sense - and hence that the distribution produced by Direct Interface and CVCV is likely to reflect some linguistic reality.

2. Interface Dualism in SPE

(2) the phonological cycle

(transformational cycle) - later known as Phase theory (Chomsky 2001, Distributed Morphology)

actually, earliest version: Chomsky et al. (1956:75)

- a. the cycle is the general principle of phonological interpretation in grammar that applies within the word (morphology) as much as among words (syntax)

"Investigation of English and other languages confirms this expectation and permits us to formulate the principle of the transformational cycle in full generality, applying to all surface structure whether internal or external to the word." Chomsky & Halle (1968:27)

- b. cyclic interpretation of the string that is submitted to phonology proceeds step by step from inner domains outwards.

"Regarding a surface structure as a labeled bracketing [...], we assume as a general principle that the phonological rules first apply to the maximal strings that contain no brackets, and that after all relevant have applied, the innermost brackets are erased; the rules then reapply to maximal strings containing no brackets, and again innermost brackets are erased after this application; and so on, until the maximal domain of phonological processes is reached." Chomsky & Halle (1968:15)

- c. the phonological cycle is natural and intuitive:

"Notice, once again, the that the principle of the transformational cycle is a very natural one. What it asserts, intuitively, is that the form of a complex expression is determined by a fixed set of processes that take account of the form of its parts." Chomsky & Halle (1968:20)

(3) **boundaries**

#, +, =

- a. # is "syntactic"
- b. + is "morphological"
- c. = only occurs in learned vocabulary:
per=mit, de=signate, con=de=scend and the like (Chomsky & Halle 1968:94s)

(4) interplay of the phonological cycle and boundaries

- a. SPE was aware of the fact that English has two distinct classes of affixes:

"Alongside of the affixes that affect stress placement [...], there are other 'neutral affixes' which characteristically play no role in the placement of stress, for example, the adjective-forming affixes -y, -like, -able, -ish and affixes such as -ing, -past tense, -hood, -ness, -ly, -wise. We can indicate the fact that an affix is neutral by making use of the # boundary." Chomsky & Halle (1968:84)

- b. the Main Stress Rule shifts stress one vowel right when an affix is added.
- c. hence the challenge: the Rule must not apply to strings that contain stress-neutral affixes.
- d. solution:
 - 1. stress-neutral affixes come with a # boundary
 - 2. the Main Stress Rule is amended so that it is blocked by #
- e. párent - parént-al vs. párent-hood
[parent] → párent
[[parent] al] → [[párent] al] → [parént al]
[[parent] #hood] → [[párent] #hood] → [párent #hood]

(5) summary

- a. SPE encodes extra-phonological information
 - 1. procedurally - the phonological cycle
 - 2. representationally - boundaries

- b.

this is what I call Interface Dualism .
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- c. Interface Dualism is an important measure of interface theories.
Later theories will try to maximise or to eliminate either way of transferring extra-phonological information to the phonology:
 - 1. all-is-procedural: Lexical Phonology, Distributed Morphology (for the time being)
 - 2. all-is-representational: Prosodic Phonology

==> these attempts have failed.
- d. SPE was right (although not explicit):
 - 1. any interface theory must have both ways of transmission
 - 2. attempts at monoculture are doomed to failure.

3. The KK-classification and what it misses

- (6) Kenstowicz & Kisseberth (1977:83ss, 1979:407ss)
propose a classification according to effect of extra-phonological information on phonological processes:
given two morphemes M_1 and M_2 , their concatenation may
- have no effect at all:**
phonology works as if there were no morpho-syntactic division, i.e. as if the sequence of sounds were monomorphemic.
 - block**
a process that would apply if the morpho-syntactic division were not there.
 - trigger**
the application of a process that would not take place if the morpho-syntactic division were not there. These cases are known as **derived environment effects**.
- (7) stress-related interface phenomena correspond to neither of the KK-classes:
- no doubt that párent - parént-al vs. párent-hood is an interface phenomenon: the boundary has a phonological effect.
 - however, it does neither block a phonological process, nor is it a condition on a phonological event.
 - ==> the stress-assigning process always goes into effect no matter what.
 - ==> morpho-syntax does not define **whether or not** the process applies - it defines **how** it applies.
- (8) final 3-way classification of interface processes
- those that modify the applicability of phonological processes
 - Type A: blocking:**
a specific concatenation blocks a process that otherwise goes into effect.
 - Type B: triggering**
a specific concatenation is a condition on a process which otherwise does not go into effect.
 - Type C: modifying**
those that modify the application of a phonological process
e.g. stress assignment: párent - paréntal vs. párent-hood
- (9) illustration
- Type A (blocking)
 - Trisyllabic Shortening** (or Laxening)
 - the long vowel or diphthong in bisyllabic words is shortened if a class 1 suffix is added, but not if a class 2 suffix is added:
class 1: sane [sejn] - san-ity [sænitɪ]
Christ [kraɪst] - Christ-ian [krɪstʃən]
class 2: maiden [meɪdən] - maiden-hood [meɪdənˈhʊd]
wild [waɪld] - [waɪldnəs]
 - quite some counter-examples exist:
obese [əʊbiːs] – obese-ness [əʊbiːsnəs], but obes-ity [əʊbiːsɪtɪ]
wild [waɪld] - [waɪldnəs], but wilderness [wɪldənəs]

2. **nasal assimilation**
 - im-possible, i[ŋ]-credible vs.
u[ŋ]-predictable, u[ŋ]-comfortable
 - nasal assimilation is blocked with class 2 un- (but not with class 1 in).
- b. Type B (triggering)
 1. **Polish palatalisation** (Rubach & Booij 1984:3ss)
 - across a morpheme boundary:
grymas - gryma[ɕ-ɛ] "grimace, id. LOCsg"
głód - gło[d̪ɔ-i]ć "hunger, to grouch, to starve sb"
 - but not morpheme-internally
[dɛ]sant, [di]nosaur, pro[tɛ]st "landing, dinosaur, protest"
 2. **nasal cluster simplification**
(Mohan 1986:21ss, Halle & Mohanan 1985:95s, Borowsky 1986:232ss)
 - | | | | |
|----------|--------------|-----|-----------------|
| # | class 2 | | class 1 |
| sign [n] | sign-ing [n] | vs. | sign-ature [gn] |
| damn [m] | damn-ing [m] | vs. | damn-ation [mn] |
 - g/n must be underlyingly present, hence there must be g/n-deletion
deletion is triggered by class 2 suffixes
 3. **English angma: g-deletion**
Halle & Mohanan (1985:62ss)
 - | | | | |
|----------|--------------|----------------|------------------------------|
| # | class 2 | morph-internal | class 1 |
| sing [ŋ] | sing-er [ŋ] | finger [ŋg] | young-er [ŋg] |
| | sing-ing [ŋ] | | long-er [ŋg] vs. long-er [ŋ] |
- c. Type C (modifying)
stress-neutral vs. stress-shifting affixes in English

bare stem	class 1		class 2
párent	parént-al	vs.	párent-hood
válid	valid-ity	vs.	válid-ness
átom	atóm-ic	vs.	átom-ise

4. How interface theories manage the 3-way typology

- (10) distribution of procedural and representational management over the three types of interface phenomena in various theories
[there is no time to make a demonstration for the particular distribution of each theory
- see Scheer (2006) (handout available online) for that]

		Type A (blocking)		Type B (triggering)		Type C (modifying)	
		proced.	repres.	proced.	repres.	proced.	repres.
a.	SPE	—	#	—	#	cycle	#
b.	Lex Phon	level 1	—	level 2	bracket	level 1	—
c.	Pros Phon	—	Pros Hier	—	Pros Hier	?	
d.	Kaye (1995)	—	domain edges	does not exist		cycle & robustness	—
e.	Distr Morph	(phase?)	# (?)	?		phase	—
f.	Stratal OT	level 1	Pros Hier	lexicalised		level 1	—
g.	Direct Interf.	—	yes	—	yes	phase & robustness	—

- (11) observations
- a. Type A (blocking) is managed representationally by all theories save Lexical Phonology.
 - b. Type C (modifying) is managed procedurally by all theories.
 - c. no theory except SPE makes a representational device intervene for Type C (modifying).
 - d. Type B (triggering) is the most "irregular" one: interface theories seem to avoid it and propose non-phonological accounts.
 - e. only Lexical Phonology makes a procedural device intervene in Type B (triggering).
 - f. Direct Interface, to be introduced below, distinguishes between those phenomena that block or trigger the application of rules (Type A+B) and those that modify it (Type C):
 1. Type A+B are due to representational intervention - no procedural element
 2. Type C is due to procedural intervention - no representational element

==> this is the take of SPE, except that there is no representational contribution to Type C.
- (12) correlation between segmental vs. suprasegmental phenomena and interface Types
- a. Type C (modifying) = stress
it appears that ALL Type C phenomena concern stress placement
==> or am I missing something?
 - b. there is debate regarding segmental phenomena:
 - ALL theories have a representational management
 - some theories in addition have a procedural management
 - c. Direct Interface makes the distribution complementary:
 - suprasegmental = procedural
 - segmental = representational
 - d. this division coincides with the traditional notion of **sandhi**:
 1. only segmental phenomena can be sandhi since sandhi is strictly **local**: something happens at the seam of two morphemes or words.
 2. stress placement does not modify the local environment of a particular morpho-syntactic division and therefore could not be sandhi.
 - e.

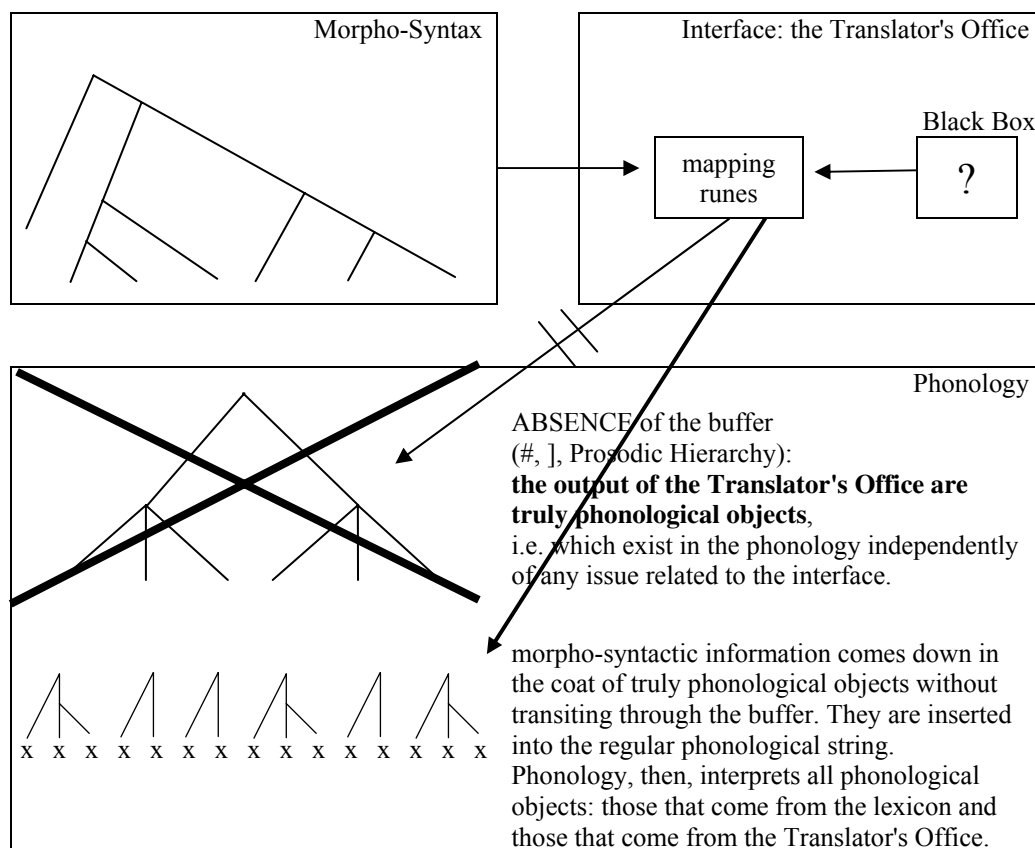
Direct Interface and CVCV enforce a purely representational management of phenomena that are
1. local = 2. segmental = 3. sandhi

5. Direct Interface

- (13) principles of Direct Interface
Scheer (forth a), Scheer (forth b)
- a. Interface Dualism
SPE was basically right - only that boundaries must not be diacritic
 - b. a Translator's Office in modular no-man's land
 - c. privacy
only domestic phonological law rules
 1. morpheme-internally anyway
 2. at morpheme edges as long as there is no higher intervention
 - d. representational intervention is LOCAL (sandhi)

- e. Direct Interface is theory-neutral: it sets a frame for the interface which may then be filled in by any specific phonological theory
- f. its name comes from the **absence of any mediation**/ buffer/ sponge:
 - 1. unlike all other interface theories (except Direct Syntax), it does not impose any interface vocabulary.
 - 2. the procedural device by definition has no vocabulary: it can only make submission-decisions
 - 3. the vocabulary of the representational device is the vocabulary of the phonological theory:
==> only truly phonological objects can be the output of the Translator's Office
 - 4. this is how "non-diacritic boundaries" are achieved:
higher level intervention is both local and non-diacritic.

(14) general architecture of Prosodic Phonology



(15) actions that the Translator's Office may take

- a. representational
 - it may send down an object to the phonology. This object is a truly phonological unit and will be inserted at the boundary of two morphemes or words. Its action therefore is only local.
- b. procedural: **chunk submission**
 - it may decide to submit only a chunk of the total linear string to the phonology, and to repeat this action several times with variously sized chunks (from smaller to bigger, climbing up the morpho-syntactic tree). Every time phonology receives some chunk, it assigns phonological interpretation to it. Obviously, chunk-submission is not local in character.

- (16) hence Interface Dualism
- a. procedural: defines
 1. how much of the string is submitted
 2. how many times
 - b. representational: defines
 1. how extra-phonological information is shipped into the phonology
 2. where precisely this information lands
 3. what it may consist of
- (17) No mediation - hence no diacritics
- a. the output of the Translator's Office are only truly phonological units.
 - b. definition of "truly phonological object"
a truly phonological object is a unit that is needed for the purpose of domestic phonology and in absence of any issue related to extra-phonological information.
 - c. hence, are ruled out:
 - omegas, the Prosodic Hierarchy
 - boundaries
 - brackets
- (18) phonological theories may be evaluated according to their behaviour at the interface
- a. unlike all other theories (except Direct Syntax), Direct Interface does not impose any interface vocabulary.
 - b. its representational vocabulary are the units of whatever phonological theory is used.
 - c. different domestic phonological theories have different vocabulary, and hence make different predictions how the interface works and what is a possible interface event.
 - d. hence they may be evaluated according to their behaviour at the interface.
 - e. structuralism and SPE did exactly the right thing:
 1. here is what my phonological theory looks like
 2. hence my interface vocabulary will be one of my phonological units
 - phonemes in structuralism
 - segments in SPE
 3. this predicts absurd interface events: # → p etc.
 4. conclusion: the phonological theory must be wrong.
 - The objects which it offers for the incarnation of higher level information are the wrong ones.
 - ==> phonological theory must have other objects, of a kind that have not been discovered yet.
 - ==> autosegmental representations could have been discovered on these grounds.
- (19) dummy phonology
- a. nothing fancy is going on in phonology:
no concatenation, no ordered application of processes, no reranking of constraints – just interpretation
 - b. The only thing that phonology is able to do is to interpret whatever is stuck into its space.

c. **phonology is a chemical bath**

This may be compared to a chemical process that occurs when a piece of composite material is dipped into a chemical bath: based on the ingredients of the piece and the properties of the bath, a chemical reaction goes into effect which ennobles the original piece and makes it apt for its life in the real world. The architect of the ingredients of the piece, of its size and of the timing of its dipping is the Translator's Office. It cannot alter the pieces that come from the (morpho-syntactic) factory, but it may rearrange them, add some ingredients at given locations (morpheme and word boundaries), and it decides which piece is dipped into the bath, and how many times this repeated.

(20) Serialism is a property of the Translator's Office, not of phonology

- a. the OT-debate regarding derivationalism is perhaps vacuous:
yes, phonological interpretation is procedural
but proceduralism is not run by the phonology - it is run in the Translator's Office
- b. morpho-syntax needs a procedural interface mechanism anyway: derivation by Phase
hence doing the procedural job another time in phonology is redundant and pointless.
- c. ==> interactionism, but also OTed versions of level ordering (Stratal OT, DOT) do the job a second time by constraint reranking.
- d. this is exactly the way Kaye (1995) works:
 1. the ϕ function (= phonology) is non-procedural
 2. it applies several times to several domains (the phonological cycle)

(21) Restrictions on representational intervention I

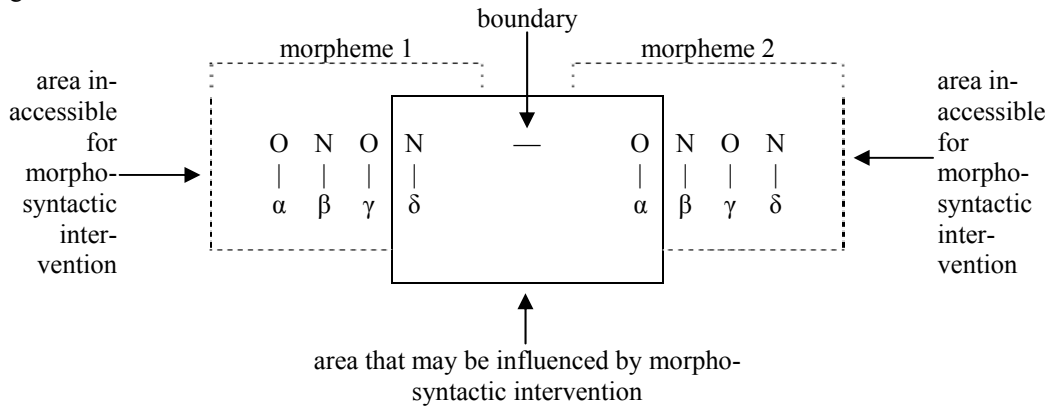
Local intervention

- a. only objects that are adjacent to the morpho-syntactic boundary at hand can be targeted.
- b. but what exactly does "adjacent" mean ?
- c. possible intervention on three objects:
 1. the last item of morpheme one
 2. the first item of morpheme two
 3. the empty space that lies between them
- d. "last item" and "first item" are notions that depend on the phonological theory used (it may be syllables, feet, x-slots etc.).
Below I use an all-purpose autosegmental representation that is as unmarked as possible.
- e. minimal requirement: the existence of domestic phonological representations.
[something that is not so obvious in some versions of OT]
- f. "first phonological item" of a morpheme
= its first syllabic constituent together with all melodic material that it dominates.
"last phonological item" of a morpheme
= its last syllabic constituent together with all melodic material that it dominates.
- g. hence no higher level intervention in the middle of morphemes
 1. a lesson from structuralism (cf. above)
 2. edge-interior asymmetry that every phonologist is aware of

- h. the **edge-interior asymmetry** is due to locality of higher intervention
1. phonological law alone governs the phonological behaviour inside morphemes.
 2. both regular domestic phonological law and extra-phonological law (i.e. the representational output of the Translator's Office) determine the phonological behaviour of morpheme edges.

In case of conflict, the extra-phonological law "wins".

higher level intervention is local: definition of its zone of influence I

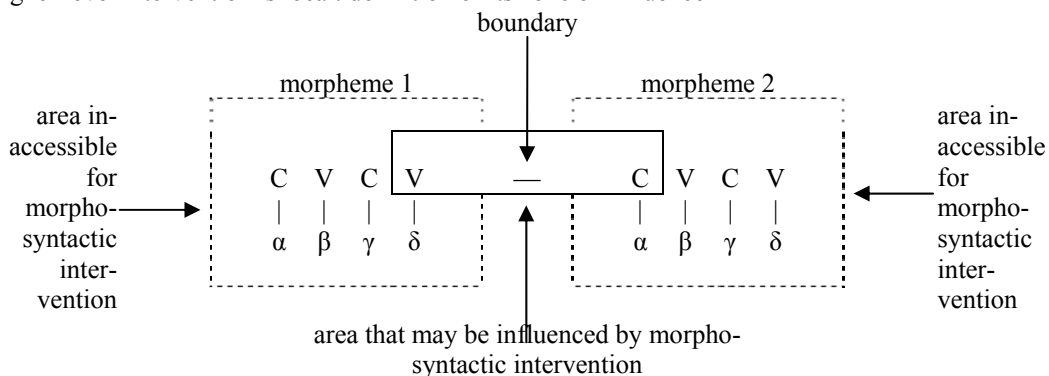


(22) Restrictions on representational intervention II

melody is invisible for morpho-syntax

- a. higher levels cannot intervene below the skeleton, i.e. manipulate melody.
- b. this is Phonology-free Syntax in the opposite direction
- c. empirical basis
 1. phonotactic and suprasegmental effects are plethoric at the edge of morphemes: extrasyllabicity, extraprosodicity, the restriction of word-initial clusters to obstruent-sonorant sequences, the allowance for heavy clusters at the right edge (e.g. English *sixths* [sɪksθs]) etc.
 2. no melodic influence on record: $p \rightarrow l / _ \#$ is unheard of.

higher level intervention is local: definition of its zone of influence II

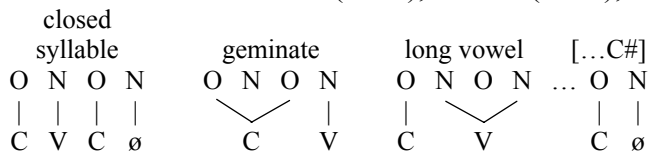


6. Direct means "your interface analysis is dictated by your phonological theory"

- (23) the direct effect: the interface analysis is dictated by the phonological theory
Hence the following recipe for analysing interface phenomena:
- identify the properties of the alternation: what is the underlying, what the surface form?
 - which are the ingredients present in the domestic phonology (i.e. from the lexicon)?
 - those that are missing in order to compute the alternation must be provided by the Translator's Office.
These may be procedural or representational.

(24) Direct Interface in CVCV

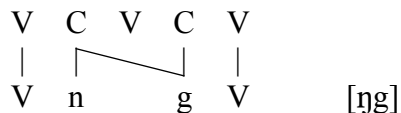
CVCV: Lowenstamm (1996), Scheer (2004), Szigetvári (1999), Cyran (2003)



(25) example: Type B (triggering)

sing - singer [ŋ] vs. finger - younger [ŋg]

- /ng/ must be underlying. Morpheme-internally, we are sure that phonology rules alone. Hence [ŋg] is the regular situation.
- what is a homorganic NC-cluster?
It is a partial geminate, i.e. a geminate whose first position is also occupied by a nasal element (Harris 1994a:69,174s)

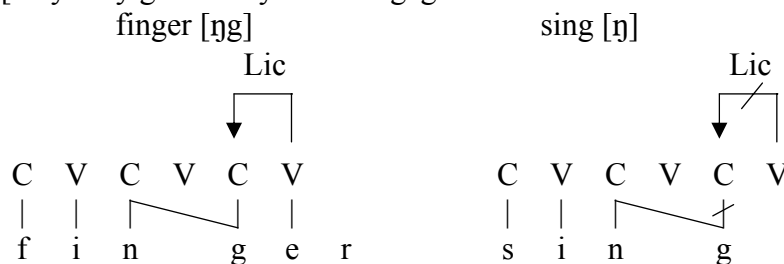


- word-finally, g stands before a Final Empty Nucleus (FEN).
- we know independently that Final Empty Nuclei (FEN) are unable to license: C# are not extrasyllabic, they behave like internal Codas: r-vocalisation, dark l.

FEN able to license = C# extrasyllabic in classical analysis

FEN unable to license = behaves like internal empty Nuclei = C# not extrasyllabic

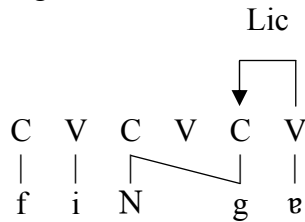
- hence g in an /ng/ cluster drops **because** it fails to be licensed.
[why only g and only when engaged in a cluster of course is a different question]



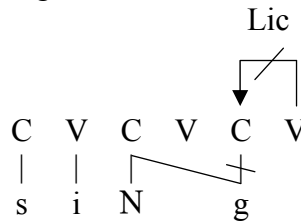
- f. **==> g is deleted when it fails to be licensed.**
- g. the two cases that imply morphology follow:
 1. sing-er, sing-ing etc.
level 2 suffixes do not carry any Interface order: phonology rules alone.
 2. young-er
level 1 suffixes come with an Interface order: "you FEN preceding the level 2-boundary are a good licenser".

illustration

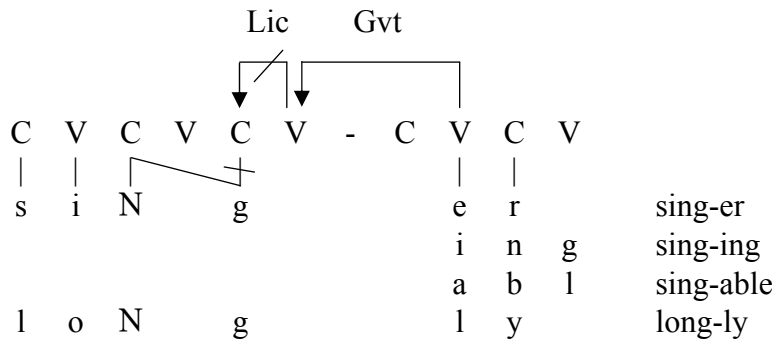
finger



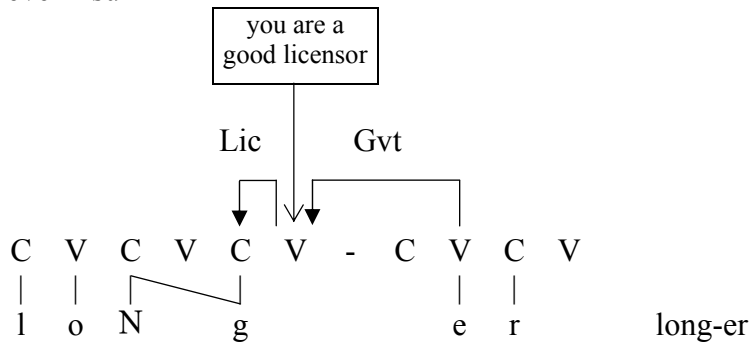
sing



level 2 suffix



level 1 suffix



(26) summary

- a. Direct Interface and CVCV **enforce** a representational management of Type B (triggering) phenomena.
- b. that is, the Translator's Office intervenes when a class 1 suffix is added: it makes the preceding FEN a good licenser (although it is not qua phonology).
- c. ==> complementary distribution of interface phenomena and interface actions:
 1. Type A (blocking) and Type B (triggering) are representational - and **only** representational
 2. Type C (modifying) is procedural - and **only** procedural

(27) conclusion:

- a. procedural intervention only occurs for the management of stress
- b. does this make any sense?
- c. yes. Reasons:
 1. the complementary distribution itself

2.

representational = segmental = local = sandhi vs. procedural = suprasegmental = non-local
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3. segmental vs. suprasegmental is a fundamental division in phonology:
below vs. above the skeleton
e.g. McMahon (2003), de Lacy (2002), Scheer (2004:§216)
it is therefore plausible that this opposition also defines whether the Translator's Office uses representational or procedural intervention.
4. stress is non-local by definition, while segmental effects always have a local source.
==> Dresher (1996:48)
5. representational = local
vs.
procedural = non-local
makes sense: an object that comes down on the phonology necessarily has a local landing site. It couldn't be in different locations at the same time, hence it can only have a local effect.
6. stress could not possibly be a Type A or Type B phenomenon since it is never blocked or triggered: a word without stress does not exist.
Only segmental processes can be blocked or triggered.
Hence if stress is to be manipulated by extra-phonological information, the only possible means is procedural.
==> consequence in SPE-type **Rule-Ordering exercises**: stress placement rules may be necessarily ordered with respect to others, but they NEVER bleed or feed them - reason: bleeding & feeding relies on blocking/ triggering of rules.
Students always panic when they cannot find out whether it is one of the four relationships.
7. stress has no phonological materiality:
 - it has no independent identity: it only enhances an object that already exists
 - it is not a feature
 - it is gradient (primary, secondary, ternary)
 - the question with stress is "where does it fall", while segmental effects are about the question "is there an effect or not?"
==> Dresher (1996)
8. therefore the complementarity is derived
 - stress could only be handled procedurally because it is non-local and therefore does not qualify for representational management, which is local by definition.
 - segmental effects always have a local source, and representational management is necessarily local.

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WEB: references followed by this mention are available at www.unice.fr/dsl/tobias.htm.

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